

WBC 13-Class Classification Using ConvNeXt-Tiny

Problem Statement

White Blood Cell (WBC) classification is done manually by lab technicians to diagnose diseases like leukemia, infections, and anemia. This is slow, error-prone, and subject to human variation. The goal of this project is to build a deep learning model that automatically classifies microscopy images into 13 WBC types.

Three key challenges made this non-trivial:

- **Class Imbalance:** SNE had 11,135 images but PLY had only 11. Training directly on this would make the model ignore rare classes completely.
- **Microscopy Noise:** Camera shot noise (grain) in lab images can mislead the model into learning noise patterns instead of cell morphology.
- **Background Clutter:** WBCs occupy only 20–30% of the image area. The rest is red blood cells and plasma — irrelevant background.

The 13 classes to identify:

Code	Full Name	Key Biological Feature
SNE	Segmented Neutrophil	Multi-lobed nucleus (3-5 lobes), most abundant WBC in blood
LY	Lymphocyte	Small cell, high nucleus-to-cytoplasm ratio, dense round nucleus
MO	Monocyte	Largest WBC, kidney-shaped nucleus, abundant grey cytoplasm
BL	Blast	Immature cell, fine chromatin, large nucleus — indicator of leukemia
EO	Eosinophil	Bi-lobed nucleus, large bright orange/red granules
BA	Basophil	Dark purple-black granules that obscure the nucleus
BNE	Band Neutrophil	Immature neutrophil, band-shaped unsegmented nucleus
VLY	Variant Lymphocyte	Activated lymphocyte, larger with more cytoplasm than normal LY
MY	Myelocyte	Granulocyte precursor, round nucleus, secondary granules forming
MMY	Metamyelocyte	Between myelocyte and band; kidney-shaped, indented nucleus
PMY	Promyelocyte	Large cell, prominent nucleoli, primary (azurophilic) granules
PC	Plasma Cell	Eccentric nucleus with clock-face chromatin pattern, abundant cytoplasm
PLY	Prolymphocyte	Between lymphoblast and lymphocyte; has a visible nucleolus

Methods and Pipeline

Full Pipeline Overview

Below is the complete pipeline we followed from raw images to final results. Each step is explained in detail in the sections that follow.

Step	Name	What It Does
1	Environment Setup	Install libraries, set random seed, configure GPU
2	Data Loading	Load CSV splits, map class names to indices, build file paths
3	Path Verification	Spot-check 50 random paths per split to confirm images exist on disk
4	Minority Class Augmentation	Synthetically increase rare classes (PLY, PC, PMY, VLY, BNE, MMY, MY, BA) to 600 samples each
5	Image Denoising (denoise_wbc)	3-stage pipeline: NLM denoising + Bilateral filter + CLAHE contrast enhancement
6	ROI Cropping (crop_wbc_roi)	Detect the WBC cell using 5-method cascade, crop tightly with 20% padding
7	Dataset Class + Sampler	PyTorch Dataset applies all preprocessing at runtime; WeightedRandomSampler balances batches
8	Model Architecture	ConvNeXt-Tiny pretrained on ImageNet, head replaced for 13-class output
9	Training with AMP + Early Stopping	Phase 1: Frozen backbone for 3 epochs. Phase 2: Full fine-tune at lower LR. Saves best model by validation F1.
10	3-Fold Cross Validation	StratifiedKFold ensures class balance across folds; produces OOF predictions for unbiased evaluation
11	Final Model Training	Train on full train+val combined using same hyperparameters
12	Evaluation & Visualizations	Confusion matrix, per-class F1/Precision/Recall, training curves, GradCAM, error analysis

Class Imbalance — the main data challenge

The class distribution was severely imbalanced. Below are the approximate counts in the training set:

Class	Approx Count
SNE	11,135
LY	6,886
MO	2,454
BL	1,699
EO	798
BA	420

BNE	340
MMY	335
MY	405
VLV	282
PMY	76
PC	56
PLY	11

Minority Class Augmentation

Minority classes were extremely limited.

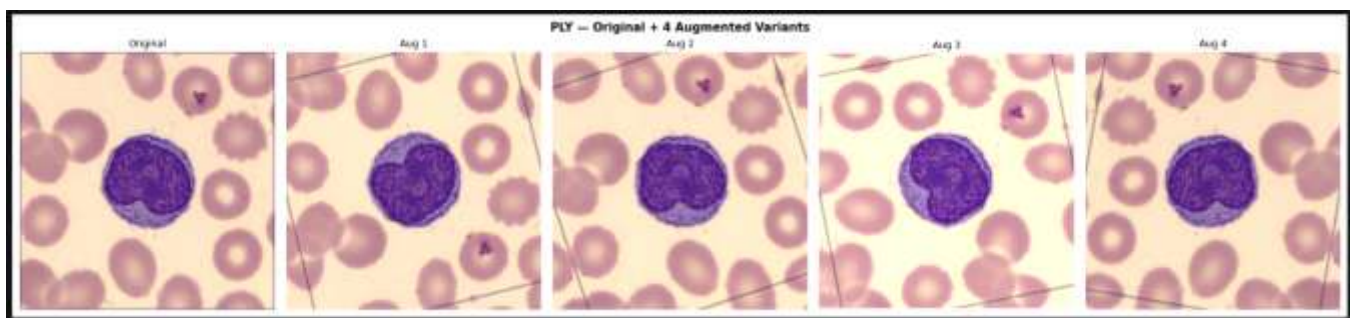
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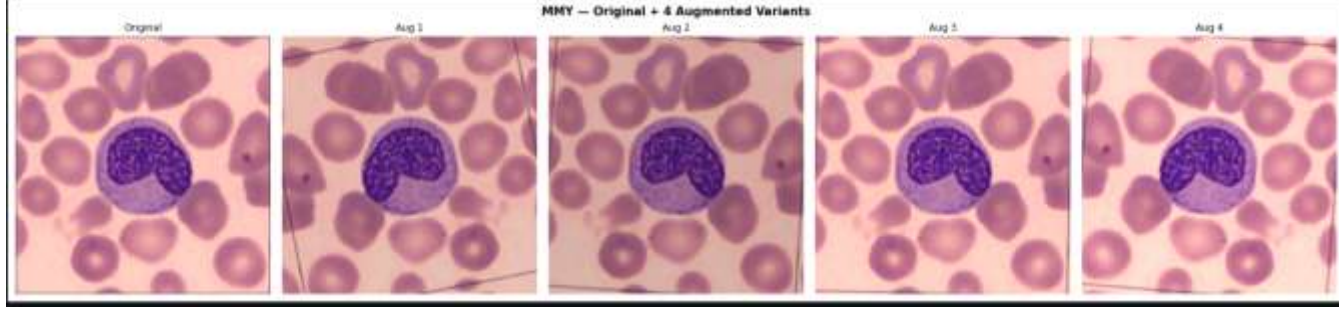
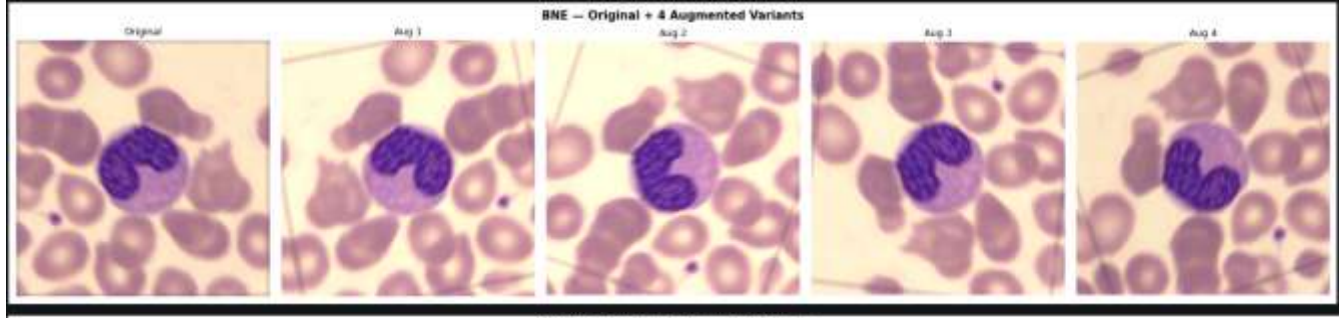
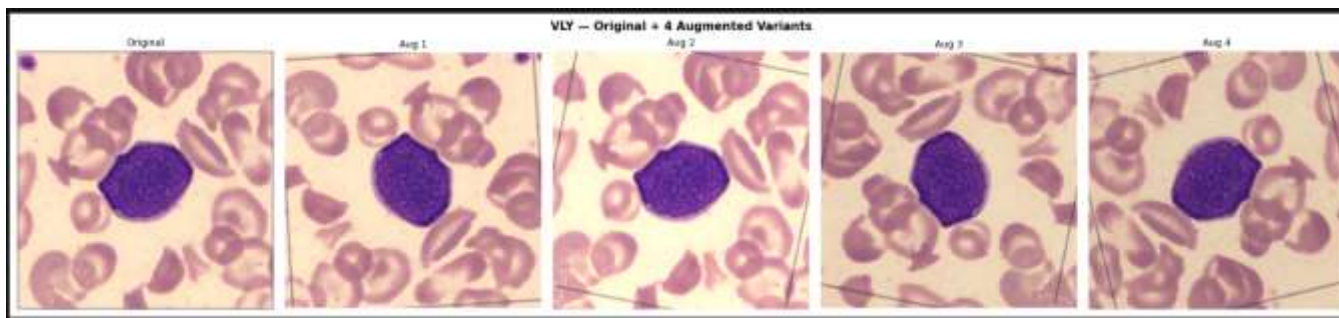
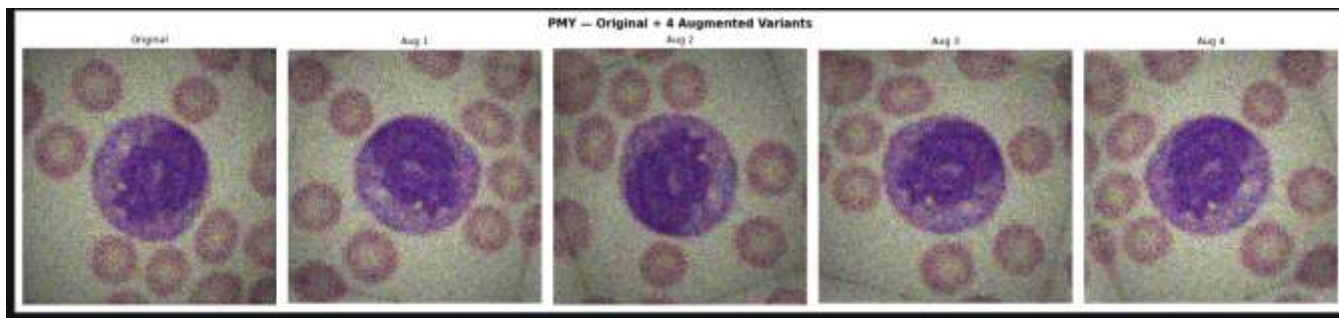
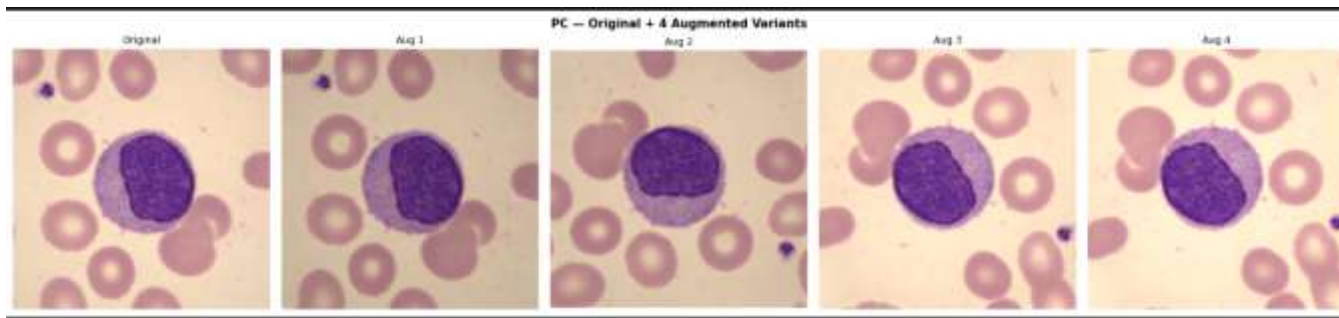
- PLY
- PC
- PMY

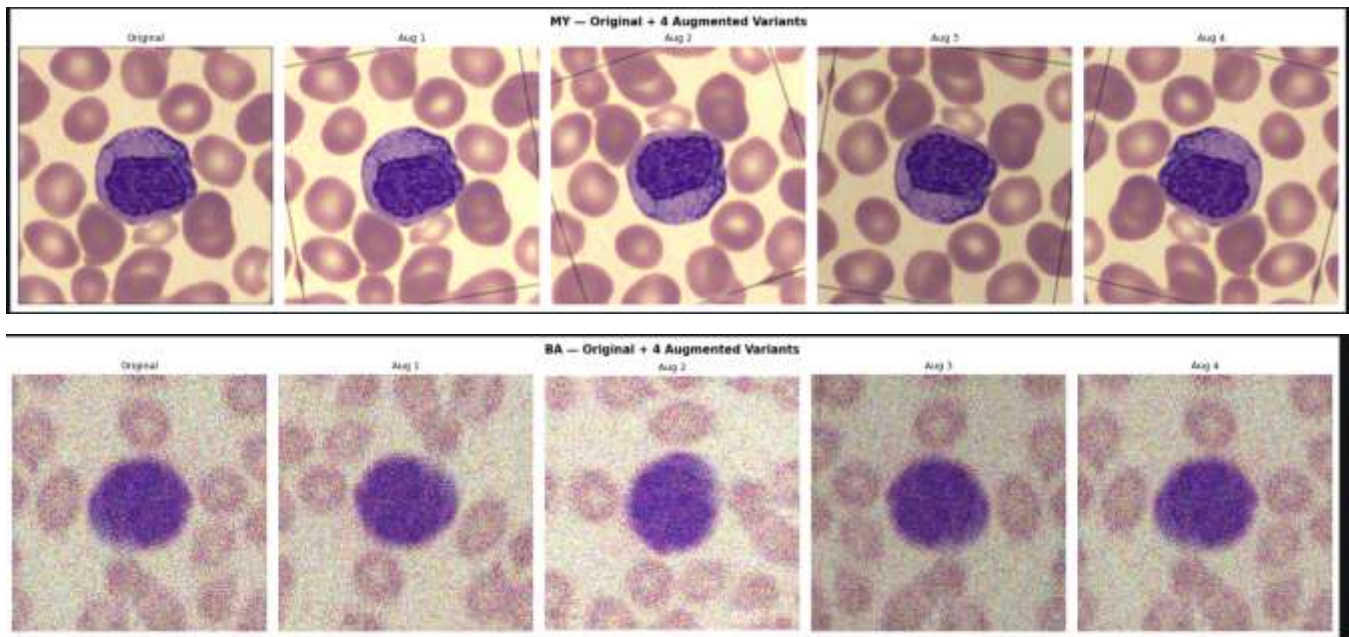
Without augmentation:

- model overfits
- minority recall collapses
- macro-F1 decreases

Class	Current Count	Target	To Generate
PLY	11	600	589
PC	56	600	544
PMY	76	600	524
VLV	282	600	318
BNE	340	600	260
MMY	335	600	265
MY	405	600	195
BA	420	600	180







After augmentation, total training samples increased from 24,897 to 27,772.

Preprocessing Visualisation

Context-Aware ROI Cropping

Why ROI Cropping Was Needed:

Raw images contain:

- large empty background
- RBC clutter
- irrelevant regions

ROI extraction focuses the model on the actual WBC.

Why Nucleus-Only ROI Was Rejected

Nucleus-only cropping removes important diagnostic structures.

Important morphological features include:

- cytoplasm
- granules
- lobes
- membrane
- chromatin
- RBC context

Some classes depend more on cytoplasm than nucleus.

Example:

Class	Important Morphology
EO	granules
BA	dense granules
PMY	cytoplasm texture
BNE	nuclear band shape
MMY	indentation morphology

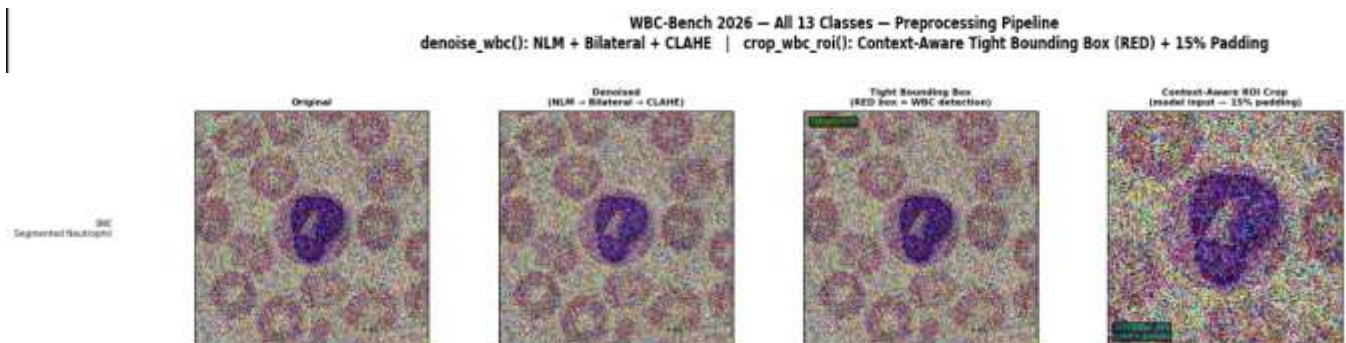
Therefore, nucleus-only ROI is biologically incorrect.

ROI Strategy

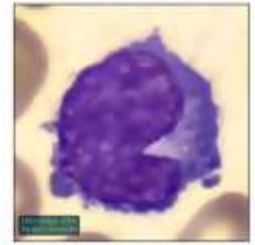
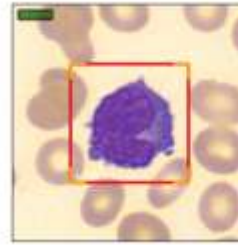
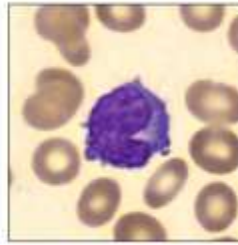
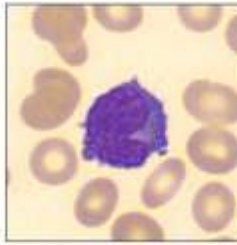
The final approach used a context-aware tight bounding box with 10–20% padding. Pipeline steps:

- RGB → HSV/LAB conversion
- WBC contour detection (5-method cascade: adaptive threshold, dark blob, Otsu, LAB, fallback)
- Largest contour extraction
- Tight bounding box extraction
- Padding addition (10–20%)
- Resize to 224×224

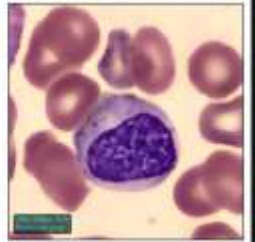
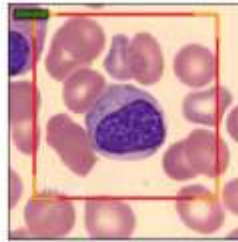
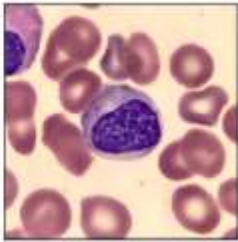
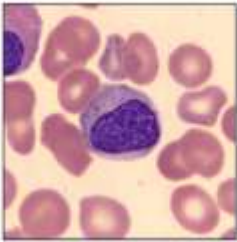
The visualization shows the full preprocessing pipeline for all 13 WBC classes. Col 3 shows the RED bounding box detection. Col 4 is what the model actually sees.



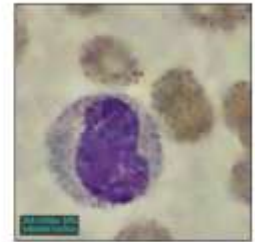
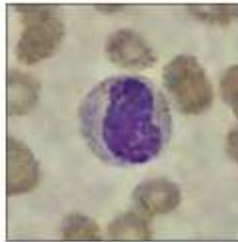
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Variant Lymphocyte



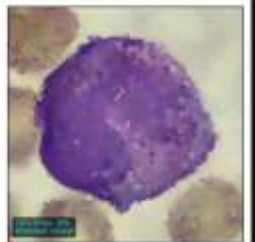
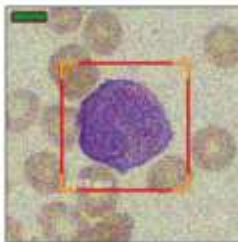
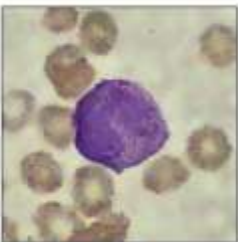
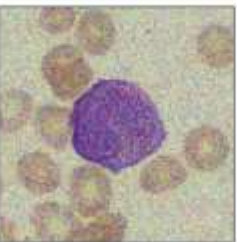
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Myelocyte



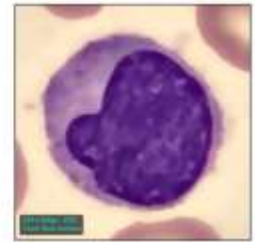
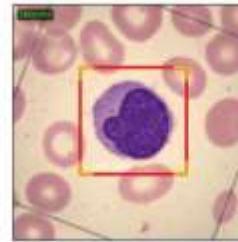
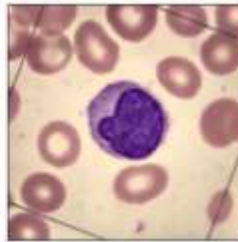
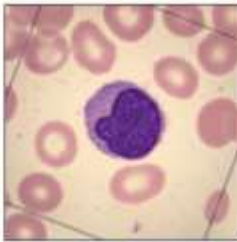
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Metamyelocyte



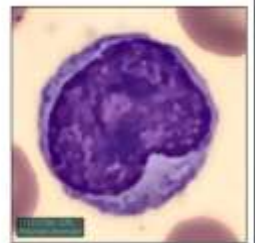
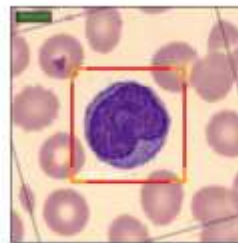
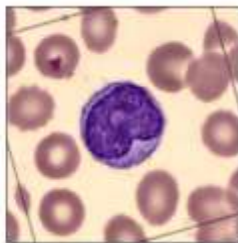
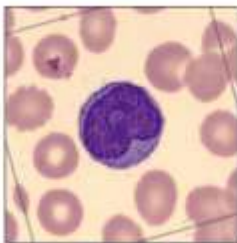
191
Polymorphocyte



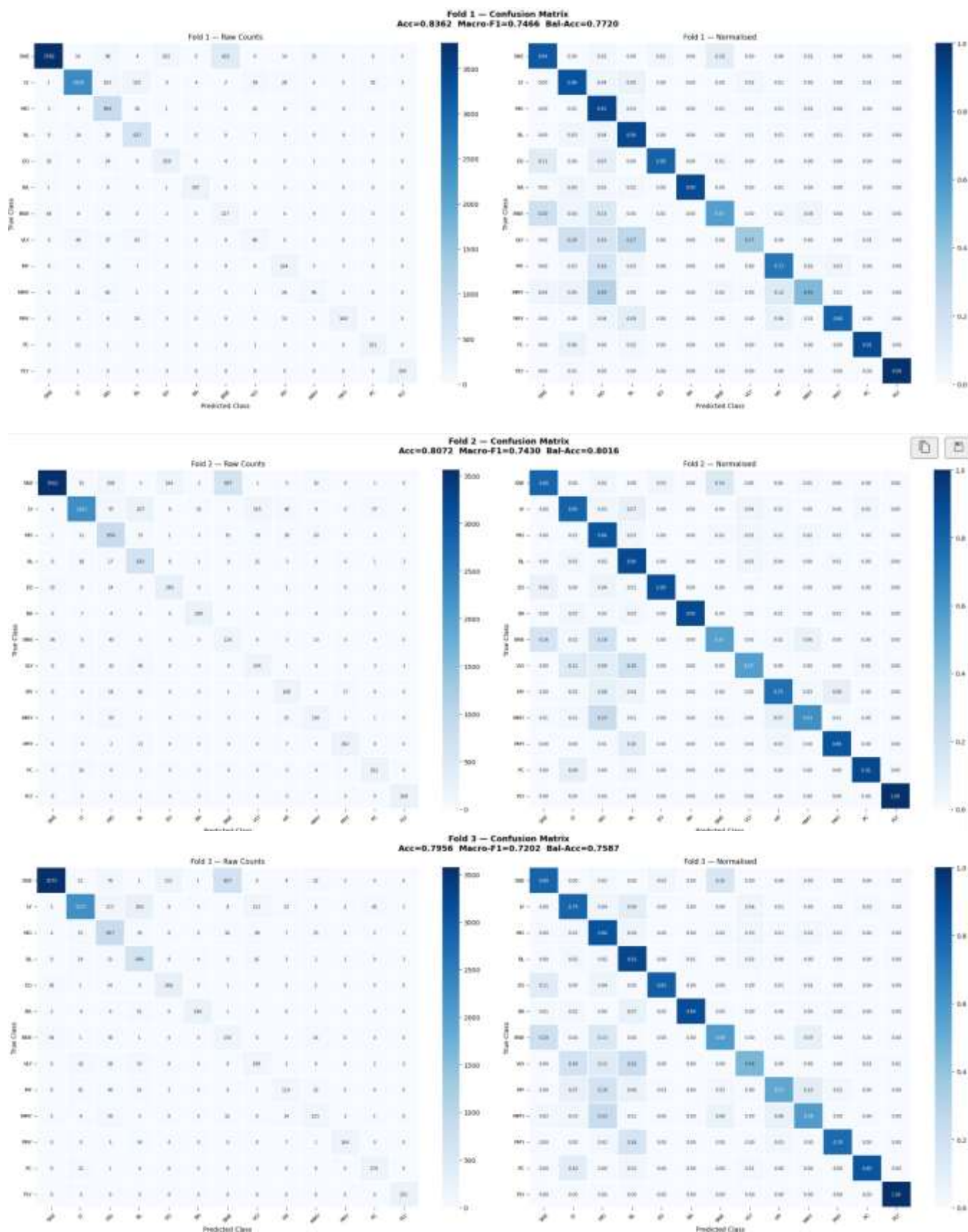
193
Plasma Cell



197
Polymorphocyte



3. Results



Cross-Validation Performance Summary

Fold	Accuracy	Balanced Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1
Fold 1	0.8362	0.7720	0.7448	0.7720	0.7466	0.8471
Fold 2	0.8072	0.8016	0.7138	0.8016	0.7430	0.8270
Fold 3	0.7956	0.7587	0.7131	0.7587	0.7202	0.8169
Mean	0.8130	0.7775	0.7239	0.7775	0.7366	0.8303

OOF (Out-of-Fold) Overall Performance

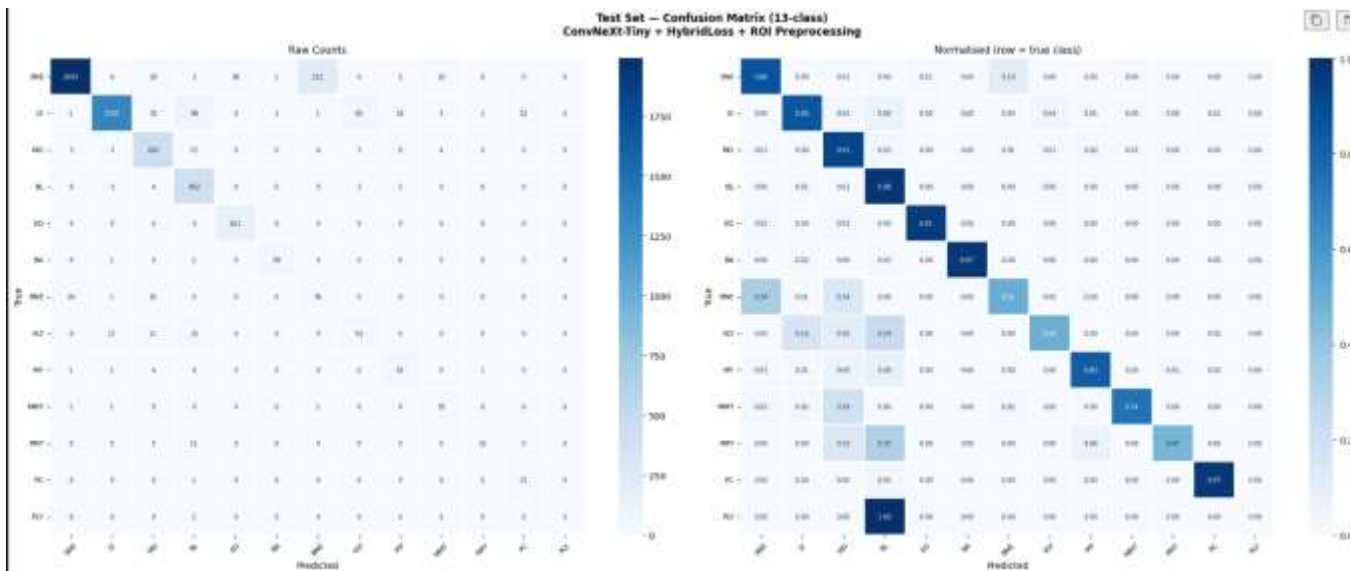
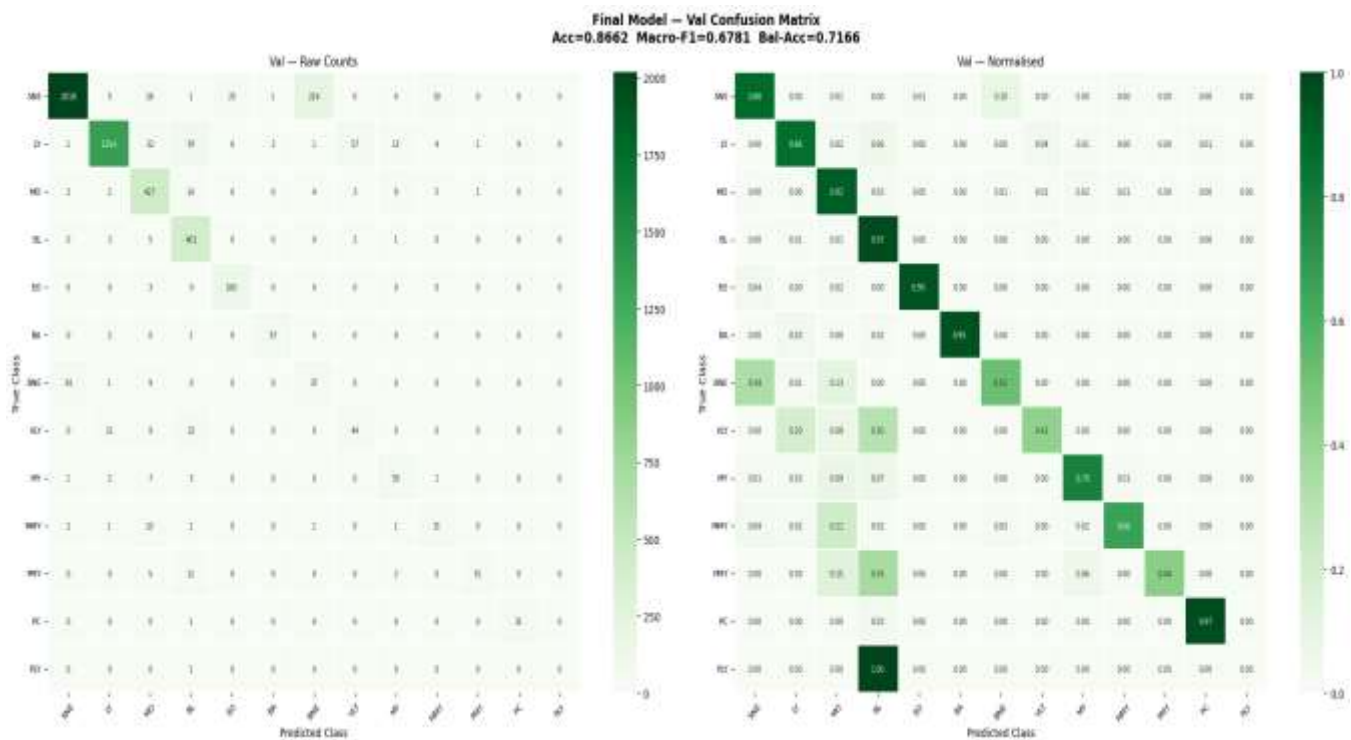
Combined OOF metrics across all 3 folds:

Metric	Score
Accuracy	0.8130
Balanced Accuracy	0.7775
Macro Precision	0.7218
Macro Recall	0.7775
Macro F1	0.7370
Weighted F1	0.8305

Per-Class Performance Metrics

Per-class metrics computed from OOF predictions (all 3 folds combined):

Class	Precision	Recall	F1 Score
SNE	0.9779	0.8127	0.8877
LY	0.9476	0.8190	0.8786
MO	0.6851	0.8784	0.7698
BL	0.6475	0.9076	0.7558
EO	0.6784	0.8376	0.7497
BA	0.9533	0.8970	0.9243
BNE	0.1770	0.5708	0.2702
VLV	0.4463	0.4646	0.4552
MY	0.5980	0.6281	0.6127
MMY	0.5961	0.5516	0.5704
PMY	0.8849	0.8122	0.8456
PC	0.7958	0.8892	0.8392
PLY	0.9694	0.9983	0.9838



Macro F1 : 0.6809

Weighted F1: 0.8773

Final Model — Validation Set Performance

The final model was trained on the full train+val set. Evaluated on the held-out validation set (5,350 images) with TTA (n=5):

Metric	Score
Accuracy	0.8662
Balanced Accuracy	0.7166
Macro Precision	0.6755
Macro Recall	0.7166
Macro F1	0.6781
Best Val Macro-F1 (training)	0.6797

Per-Class F1 — Final Model (Validation Set)

Class	F1 Score	Note
SNE	0.9257	Large sample, stable morphology
LY	0.9142	Stable lymphocyte structure
MO	0.8618	Good monocyte separation
BL	0.8200	High recall, some FP from immature cells
EO	0.9040	Distinct eosinophilic granules
BA	0.9500	Highly distinguishable dark granules
BNE	0.2189	Confused with SNE — band vs segmented nucleus overlap
VLY	0.4151	Confused with BL — immature nuclear structure similarity
MY	0.7239	Some overlap with BL fine chromatin
MMY	0.6327	Transitional stage, confused with MO
PMY	0.5882	Confused with BL — both large immature cells
PC	0.8611	Clock-face chromatin usually distinct
PLY	0.0000	Only 1 val sample; model fails to identify — expected given 11 source images

Error Analysis — Why the model confuses certain classes

Top confused pairs identified on the validation set:

True	Predicted	Count	Biological Reason
SNE	BNE	231	Lobe overlap — BNE nucleus not yet segmented, looks like early SNE
LY	BL	98	Dense nucleus confusion — both have round, dark nucleus
LY	VLY	65	Reactive morphology — VLY is activated LY with similar nucleus
SNE	EO	38	Morphological overlap
LY	MO	33	Morphological overlap
VLY	BL	25	Immature nuclear structure similarity
BNE	SNE	24	SNE multi-lobe vs BNE band shape can be ambiguous
PMY	BL	11	Promyelocyte vs blast — both are large immature cells

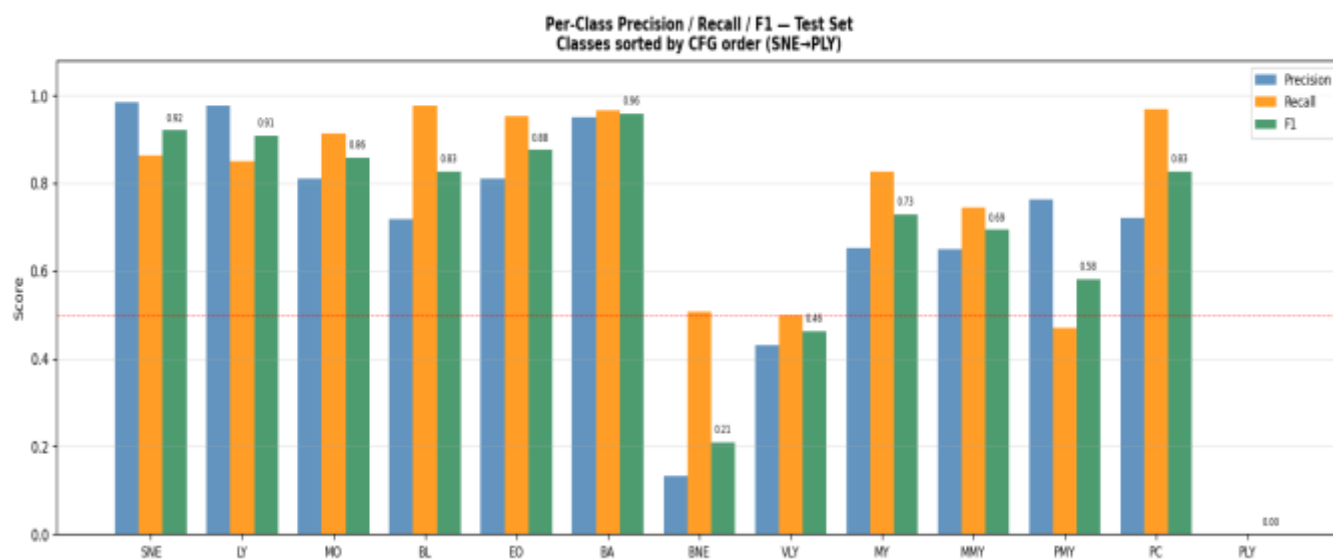
HARDEST CLASSES (lowest F1 on validation set):

Rank	Class	F1 Score	Most Confused With	Biological Reason
1	PLY	0.000	BL	Extremely rare (1 val sample); only 11 source images — model cannot generalise
2	BNE	0.209	SNE	Similar nuclear morphology — band vs segmented neutrophil boundary
3	VLY	0.415	BL	Immature appearance overlap with blast
4	PMY	0.588	BL	Both large immature cells; promyelocytes granules sometimes subtle
5	MMY	0.633	MO	Transitional maturation stage, indented nucleus similar to monocyte
6	MY	0.724	BL	Fine chromatin similarity between myelocyte and blast
7	PC	0.861	BL	Plasma cell morphology occasionally overlaps with blast

8	BL	0.820	MO	Cytoplasm/nucleus similarity between blast and monocyte
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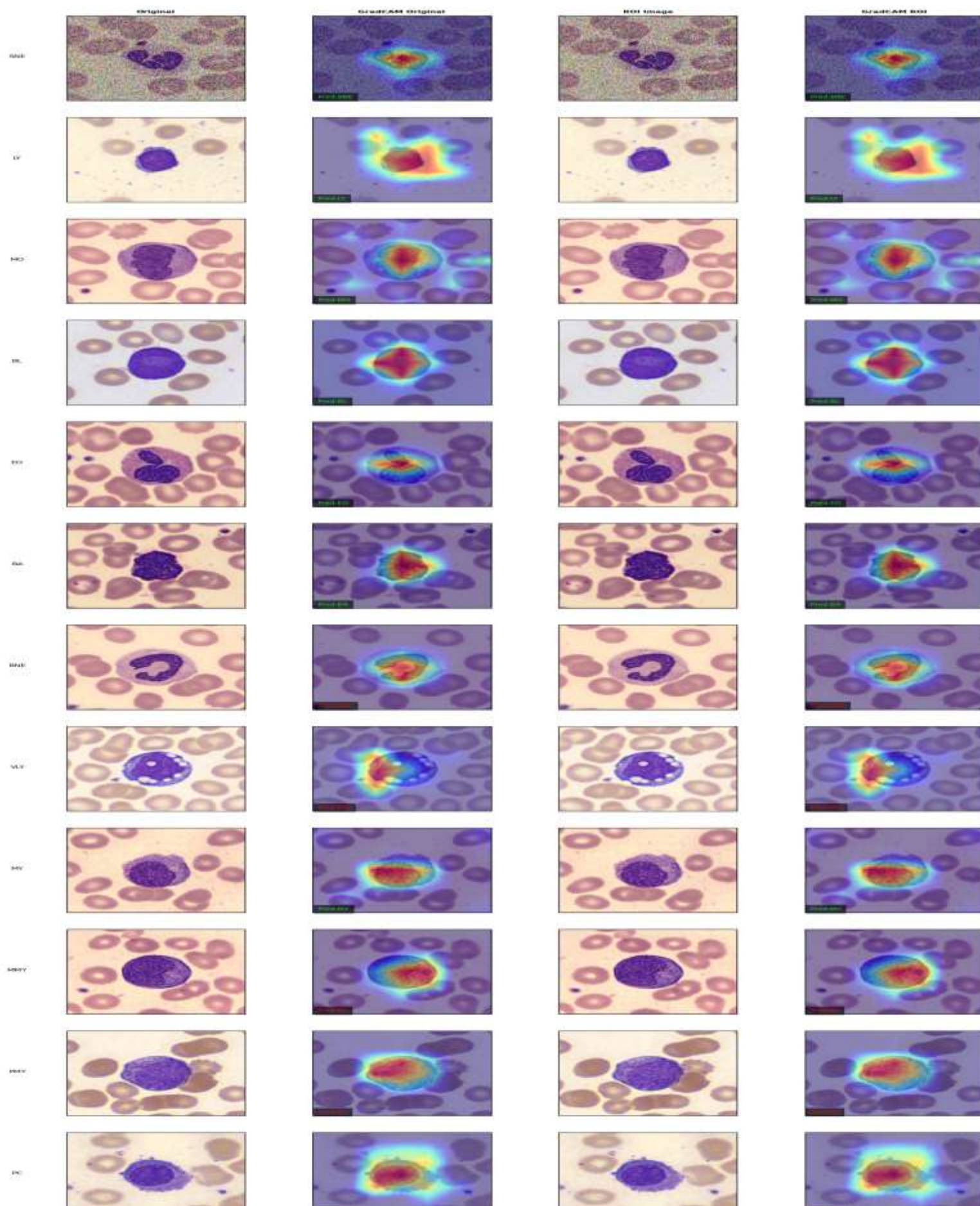
Best Performing Classes

Rank	Class	OOB F1	Val F1	Observation
1	BA	0.9243	0.9500	Highly distinguishable dark granule morphology
2	SNE	0.8877	0.9257	Large sample availability + stable multi-lobe structure
3	LY	0.8786	0.9142	Stable lymphocyte structure
4	EO	0.7497	0.9040	Distinct orange eosinophilic granules
5	MO	0.7698	0.8618	Good monocyte separation



GradCAM — What the Model Looked At

GradCAM correct predictions. Red/yellow = region the model focused on. Should highlight nucleus/granules, not background.



Comparison and Conclusion

Why We Got These Results

Design Choice	Why It Helped	Without It — What Would Happen
Minority augmentation (offline)	PLY/PC/PMY got enough training samples (600 each)	Model predicts 0 for PLY — Macro F1 collapses
WeightedRandomSampler	All 13 classes appear in every batch	Batches dominated by SNE/LY — rare classes never trained
HybridLoss (CE + Focal, 50/50)	Class-weighted focal component penalises minority misclassification more	Rare class gradients drowned out by majority classes
NLM + Bilateral + CLAHE	Model learns cell morphology, not noise	Model learns noise texture — poor generalisation
ROI crop (10–20% pad)	Model focuses only on the cell and cytoplasm	Model wastes capacity learning RBC background
Freeze 3 epochs then unfreeze	Head stabilises before backbone adapts	Noisy gradients destroy pretrained features early
Label smoothing = 0.10	Reduces overconfident wrong predictions	Model is 99% sure of wrong answer — bad calibration
Mixup augmentation (after epoch 3)	Improves generalisation, reduces overfit	Sharper decision boundaries, worse OOF performance

Known Limitations

- **MY vs MMY:** These two classes are biologically adjacent (differ only by degree of nuclear indentation). Even human experts sometimes disagree. This confusion will persist regardless of model.
- **PLY (11 real images):** All 600 augmented samples come from 11 originals. The model may not generalise to PLY images outside these 11 variations.
- **Stain quality variation:** EO vs BA confusion happens when staining is inconsistent — the orange and purple granule colours shift. A preprocessing colour normalisation step could help.
- BNE remains the second hardest class (F1 = 0.209 on val) because its band-shaped unsegmented nucleus is visually similar to early-stage SNE lobes.

Threshold Calibration

After training, per-class decision thresholds were tuned on the validation set to further improve macro-F1. Two steps were applied:

- Temperature Scaling: Best temperature $T = 0.50$ (val macro-F1 = 0.6797, same as raw argmax baseline — no improvement from temperature alone).
- Per-Class Threshold Grid Search: Each class threshold searched independently on val_df to maximise per-class F1.

Class	Tuned Threshold	Default F1	Calibrated F1	Gain
SNE	0.25	0.9197	0.9442	+0.0245
LY	0.10	0.9088	0.9331	+0.0243
MO	0.80	0.8592	0.8872	+0.0280
BL	0.85	0.8255	0.8894	+0.0639
EO	0.70	0.8750	0.9172	+0.0422
BA	0.30	0.9587	0.9672	+0.0085
BNE	0.75	0.2093	0.2558	+0.0465
VLY	0.15	0.4561	0.4981	+0.0420
MY	0.90	0.7219	0.7613	+0.0394
MMY	0.30	0.6931	0.7037	+0.0106
PMY	0.15	0.5818	0.6769	+0.0951
PC	0.80	0.8267	0.9118	+0.0851
PLY	0.40	0.0000	0.0000	0.0000

Final calibrated results on the validation set:

Metric	Score
Val Macro-F1 (after calibration)	0.7189
Val Balanced Accuracy	0.7570
Val Accuracy	0.8966
Test Predictions Generated	16,477

Conclusion

This project successfully built an end-to-end deep learning pipeline for 13-class WBC classification. The three main challenges — class imbalance, microscopy noise, and background clutter — were each addressed by a specific component: offline augmentation + weighted sampling, 3-stage denoising, and context-aware ROI cropping respectively. ConvNeXt-Tiny with frozen-then-unfrozen fine-tuning achieved a mean cross-validation macro-F1 of 0.7366 and OOF macro-F1 of 0.7370. The final model reaches 0.8662 accuracy and 0.6781 macro-F1 on the validation set. The remaining errors are biologically explainable and reflect real diagnostic challenges in haematology.